A03 – Case Study Analysis: Edge-Computing Video Analytics for Real-Time Traffic Monitoring in a Smart City

Edge-Computing Video Analytics for Real-Time Traffic Monitoring in a Smart City:

Critical Analysis of the Liverpool Smart Pedestrians Project

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Introduction and Objectives

The Liverpool Smart Pedestrians initiative seeks to tackle major issues in contemporary urban settings by facilitating real-time monitoring of pedestrians through edge-computing video analytics.

The main goal of this project is to enhance situational awareness and ensure pedestrian safety by processing video data at the edge instead of depending on centralized cloud systems. This method enables quicker decision-making in critical scenarios such as emergency evacuations and areas with high pedestrian traffic.

Urban planners encounter considerable challenges related to pedestrian overcrowding, slow data processing, and privacy issues when utilizing conventional cloud-based surveillance systems. Centralized processing frequently leads to delays and necessitates ongoing data transmission, which can be inefficient and raise ethical dilemmas. The Liverpool Smart Pedestrians project mitigates these problems by implementing intelligent sensors that perform localized analysis, thereby supporting scalable and privacy-aware smart city development.

Methodology

The project's methodology emphasizes the design, implementation, and assessment of a pedestrian sensor utilizing edge computing, which is capable of conducting real-time video analytics. This system was engineered to capture video streams, process them locally with deep learning models, and produce data for pedestrian detection and tracking without sending raw video to external servers.

Various requirements and limitations shaped the design of the system. These factors included the restricted computational resources available on embedded devices, the necessity for real-time processing, energy efficiency, and dependable performance across different environmental conditions. Additionally, preserving privacy was a vital requirement, as the system had to limit the transmission and storage of identifiable video data. These constraints influenced the choice of both hardware and software components, ensuring a balance between performance and deployability.

Technology and Implementation

The system architecture is centered around the NVIDIA Jetson TX2, an embedded edge-computing platform specifically designed for AI workloads. The Jetson TX2 was chosen for its optimal combination of GPU acceleration, power efficiency, and compact design, making it ideal for continuous use in urban settings. Its capability for on-device inference enables pedestrian detection to take place with minimal delay.

For pedestrian detection, the system employs the YOLO V3 (You Only Look Once) deep learning algorithm. YOLO V3 is particularly effective for real-time applications as it conducts object detection in a single forward pass, emphasizing speed while still achieving satisfactory accuracy. The model was developed using PyTorch, which offered flexibility for deployment and optimization. Furthermore, the SORT (Simple Online and Realtime Tracking) algorithm was utilized to monitor pedestrians across frames, allowing the system to assess pedestrian flow and movement patterns.

This implementation is consistent with the edge-computing paradigm outlined in Module 2 by decreasing dependence on cloud infrastructure, reducing bandwidth consumption, enhancing response times, and improving data privacy through localized processing.

Validation and Performance Evaluation

The validation process assessed the system's accuracy, processing speed, and resource usage.

Performance metrics comprised detection accuracy, frame processing rate, and CPU/GPU utilization on the edge device. These metrics were utilized to determine if the system could function effectively under real-world conditions.

The findings revealed that the system was able to process video streams in real time while ensuring dependable pedestrian detection accuracy. The reported frame rates suggest that the system can facilitate live monitoring scenarios without noticeable delay. Measurements of resource utilization indicated that the Jetson TX2 could handle continuous inference workloads without surpassing its operational thresholds. These findings validate the practicality of implementing AI-driven video analytics at the edge for smart city initiatives.

Real-World Applications

The project includes two real-world deployment scenarios: an indoor emergency evacuation scenario and an outdoor deployment in public areas of Liverpool. In the indoor scenario, the system was used to monitor pedestrian movement during an evacuation, demonstrating its ability to provide real-time insights in safety-critical situations. This application highlights the system's potential to support emergency response planning and crowd management.

The outdoor deployment involved monitoring pedestrian traffic in urban public spaces. This deployment demonstrated the system's usefulness for long-term urban planning by providing data on pedestrian density and movement patterns. By integrating edge sensors with gateways, the system supports distributed sensing across the city, aligning with the sensor network and edge gateway concepts covered in Module 3.

Challenges and Future Work

The project encountered numerous obstacles, including limitations in hardware, variability in environmental conditions, and concerns regarding scalability.

Embedded devices like the Jetson TX2 possess limited computational capabilities, which can restrict the complexity of models and scalability when multiple sensors are used at the same time.

Environmental elements such as lighting and weather conditions also influence the accuracy of detection.

The authors suggest that future efforts should concentrate on enhancing model efficiency and increasing system scalability.

Since 2019, progress in edge-AI hardware, including newer NVIDIA Jetson platforms, along with more efficient detection models and improved communication protocols like 5G, could greatly improve similar systems.

These advancements would facilitate higher accuracy, reduced latency, and wider deployment in smart cities.

Personal Evaluation

Overall, the Liverpool Smart Pedestrians initiative effectively showcases the real-world use of edge computing for monitoring pedestrians in real-time. The system strikes a commendable balance between performance, privacy, and scalability, serving as an excellent illustration of edge-AI implementation in smart city settings. However, a notable limitation is its reliance on a particular hardware platform, which could impact its long-term scalability and flexibility.

Future enhancements might involve optimizing models adaptively and incorporating various sensor types to bolster robustness. In spite of these constraints, the project offers a significant contribution.

to smart city initiatives by showing how edge-based video analytics can enhance urban safety and planning.

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